Tasks on K8S-07

Task 1: Execute all yaml files shown in video.

1.Toleration to pod using operator Equal:

kubectl taint nodes <node_name> special=true:NoSchedule

apiVersion: v1

kind: Pod

metadata:

  name: firstpod

spec:

  containers:

  - name: firstcontainer

    image: nginx

    imagePullPolicy: Never

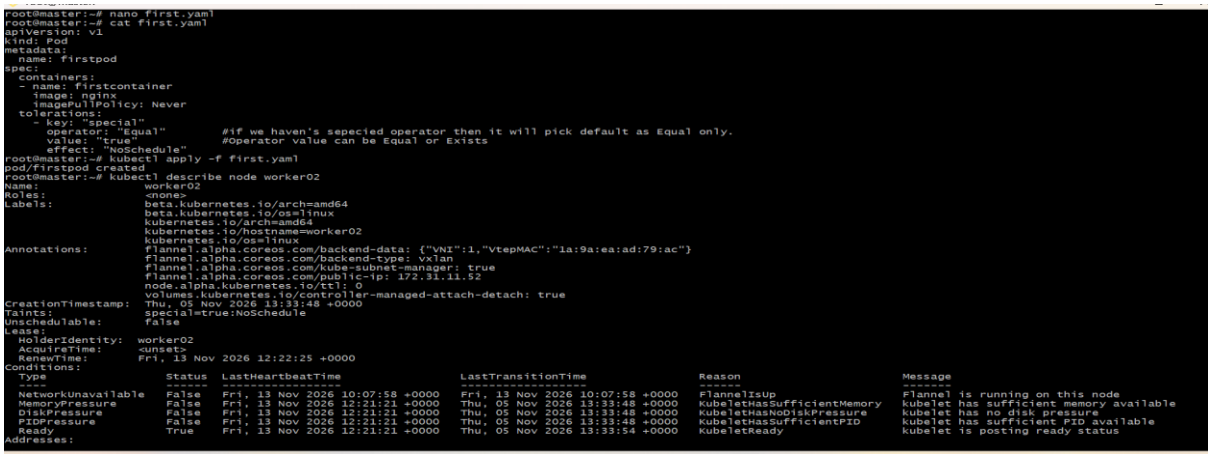
  tolerations:

    - key: "special"

      operator: "Equal" #if we haven's sepecied operator then it will pick default as Equal only.

      value: "true" #Operator value can be Equal or Exists

      effect: "NoSchedule"



2.Toleration to pod using operator Exists

kubectl taint nodes <node_name> special=false:NoSchedule

apiVersion: v1

kind: Pod

metadata:

  name: firstpod

spec:

  containers:

  - name: firstcontainer

    image: nginx

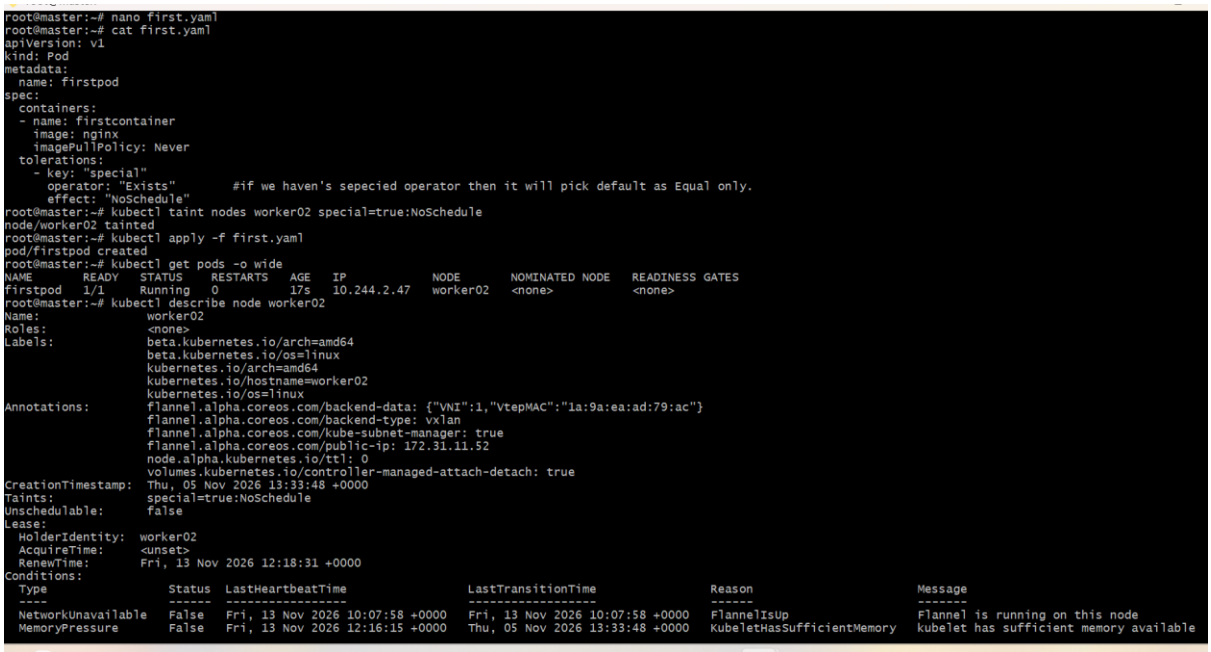
    imagePullPolicy: Never

  tolerations:

    - key: "special"

      operator: "Exists" #if we haven's sepecied operator then it will pick default as Equal only.

      effect: "NoSchedule"



3.We can use toleration seconds in tolerations of yaml if the effect is "NoExecute"

kubectl taint nodes <node_name> special=true:NoExecute

apiVersion: v1

kind: Pod

metadata:

 name: firstpod

spec:

 containers:

 - name: firstcontainer

  image: nginx

  imagePullPolicy: Never

 tolerations:

  - key: "special"

   operator: "Exists" #Operator value can be Equal or Exists

   effect: "NoExecute"

   tolerationSeconds: 60 #Pod will tolerate for 60 seconds and get terminated.

```
root@master:~# nano first.yaml
root@master:~# cat first.yaml
apiVersion: v1
kind: Pod
metadata:
  name: firstpod
spec:
  containers:
  - name: firstcontainer
    image: nginx
    imagePullPolicy: Never
  tolerations:
    - key: "special"
      operator: "Exists"          #Operator value can be Equal or Exists
      effect: "NoExecute"
      tolerationSeconds: 60
root@master:~# kubectl taint nodes worker01 special=true:NoExecute
node/worker01 tainted
root@master:~# kubectl apply -f first.yaml
pod/firstpod created
root@master:~# kubectl get pods -w
NAME       READY   STATUS    RESTARTS   AGE
firstpod   1/1     Running   0          10s
firstpod   1/1     Running   0          60s
firstpod   1/1     Terminating   0          60s
firstpod   1/1     Terminating   0          60s
firstpod   0/1     Completed     0          60s
firstpod   0/1     Completed     0          61s
firstpod   0/1     Completed     0          61s
^Croot@master:~#
```

4.Node Selector:

kubectl label nodes <node_name> env=prod

apiVersion: v1

kind: Pod

metadata:

  name: firstpod

spec:

  containers:

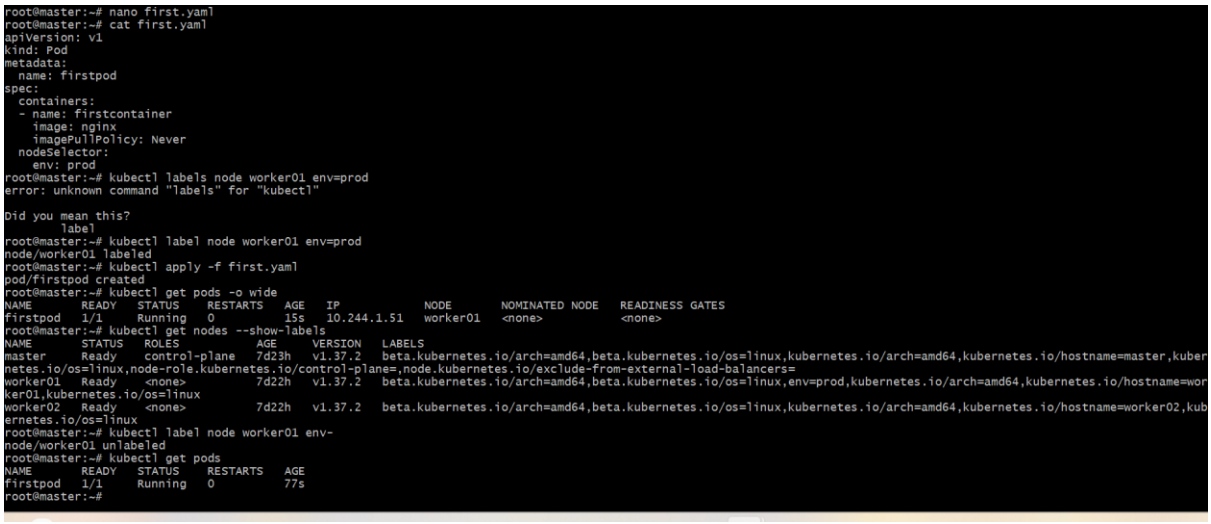
  - name: firstcontainer

    image: nginx

    imagePullPolicy: Never

  nodeSelector:

    env: prod



5. Node Affinity:

1. PreferredDuringSchedulingIgnoredDuringExecution

apiVersion: v1

kind: Pod

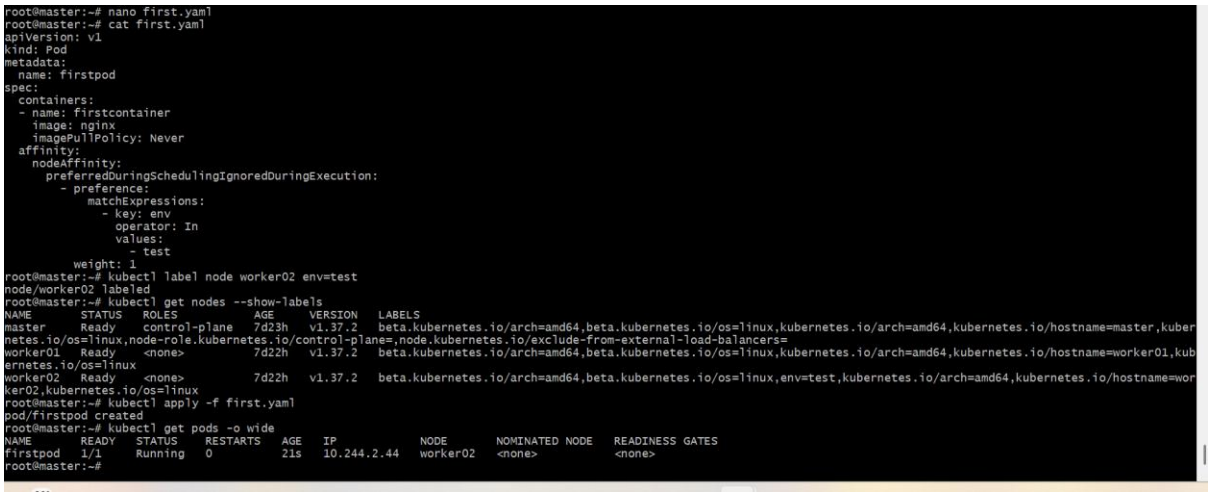
metadata:

  name: firstpod

spec:

  containers:

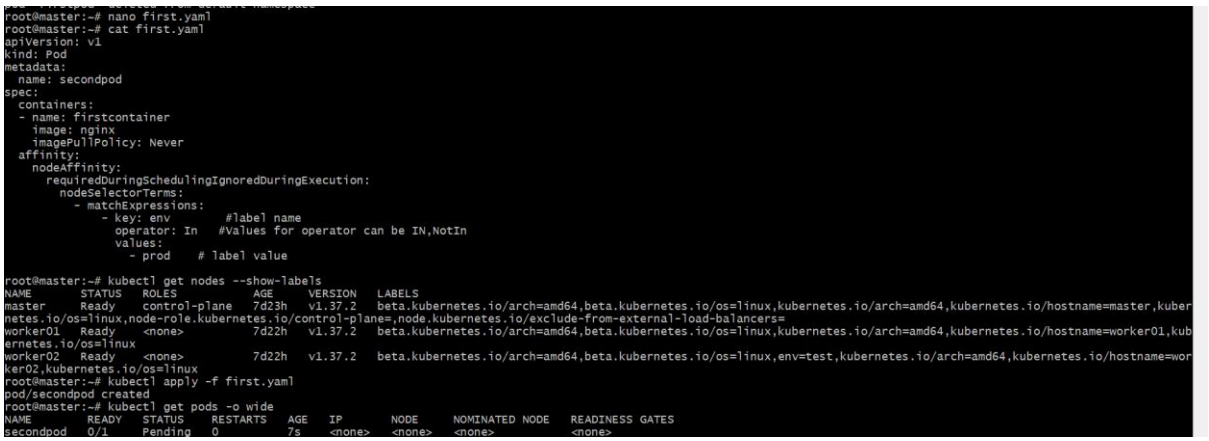
```
- name: firstcontainer
  image: nginx
  imagePullPolicy: Never
 affinity:
  nodeAffinity:
   preferredDuringSchedulingIgnoredDuringExecution:
    - preference:
       matchExpressions:
        - key: env
          operator: In
          values:
            - test
      weight: 1
```



2) RequiredDuringSchedulingIgnoredDuringExecution:

```
apiVersion: v1
kind: Pod
metadata:
  name: secondpod
spec:
  containers:
  - name: firstcontainer
```

```
  image: nginx
  imagePullPolicy: Never
 affinity:
  nodeAffinity:
    requiredDuringSchedulingIgnoredDuringExecution:
      nodeSelectorTerms:
       - matchExpressions:
          - key: env       #label name
            operator: In   #Values for operator can be IN,NotIn
            values:
              - test    # label value
```



```
root@master:~# nano first.yaml
root@master:~# cat first.yaml
apiVersion: v1
kind: Pod
metadata:
  name: secondpod
spec:
  containers:
  - name: firstcontainer
    image: nginx
    imagePullPolicy: Never
  affinity:
    nodeAffinity:
      requiredDuringSchedulingIgnoredDuringExecution:
        nodeSelectorTerms:
          - matchExpressions:
              - key: env        #label name
                operator: In   #Values for operator can be IN,NotIn
                values:
                  - prod    # label value

root@master:~# kubectl get nodes --show-labels
NAME       STATUS   ROLES           AGE     VERSION   LABELS
master     Ready    control-plane   7d23h   v1.37.2   beta.kubernetes.io/arch=amd64,beta.kubernetes.io/os=linux,kubernetes.io/arch=amd64,kubernetes.io/hostname=master,kuber
netes.io/os=linux,node-role.kubernetes.io/control-plane=,node.kubernetes.io/exclude-from-external-load-balancers=
worker01   Ready    <none>          7d22h   v1.37.2   beta.kubernetes.io/arch=amd64,beta.kubernetes.io/os=linux,kubernetes.io/arch=amd64,kubernetes.io/hostname=worker01,kub
ernetes.io/os=linux
worker02   Ready    <none>          7d22h   v1.37.2   beta.kubernetes.io/arch=amd64,beta.kubernetes.io/os=linux,env=test,kubernetes.io/arch=amd64,kubernetes.io/hostname=wor
ker02,kubernetes.io/os=linux
root@master:~# kubectl apply -f first.yaml
pod/secondpod created
root@master:~# kubectl get pods -o wide
NAME        READY   STATUS    RESTARTS   AGE   IP       NODE     NOMINATED NODE   READINESS GATES
secondpod   0/1     Pending   0          7s    <none>   <none>   <none>           <none>
```

```
root@master:~# nano first.yaml
root@master:~# cat first.yaml
apiVersion: v1
kind: Pod
metadata:
  name: thirdpod
spec:
  containers:
  - name: thirdcontainer
    image: nginx
    imagePullPolicy: Never
  affinity:
    nodeAffinity:
      requiredDuringSchedulingIgnoredDuringExecution:
        nodeSelectorTerms:
          - matchExpressions:
              - key: env        #label name
                operator: NotIn   #Values for operator can be IN,NotIn
                values:
                  - prod    # label value

root@master:~# kubectl apply -f first.yaml
pod/thirdpod created
root@master:~# kubectl get pods -o wide
NAME        READY   STATUS    RESTARTS   AGE     IP            NODE       NOMINATED NODE   READINESS GATES
secondpod   0/1     Pending   0          3m37s   <none>        <none>     <none>           <none>
thirdpod    1/1     Running   0          5s      10.244.2.45   worker02   <none>           <none>
root@master:~#
```

Task 2: Taint a Node and Schedule a Tolerant Pod Taint a node with special=true:NoSchedule. Create a pod with a toleration that matches the taint, allowing it to be scheduled on the tainted node.

Objective: Apply the taint special=true:NoSchedule to a worker node and create a Pod with a matching toleration.

- **Step 1: Apply a taint to Worker-01**
  kubectl taint nodes worker-01 special=true:NoSchedule
- **Step 2: Verify the taint**
  kubectl describe node worker-01
- Under Taints, verify:
  special=true:NoSchedule
- **Step 3: Create a Pod YAML file**
  vi tolerant-pod.yaml
- Add:

  apiVersion: v1

  kind: Pod

  metadata:

  name: tolerant-pod

  spec:

  nodeSelector:

  kubernetes.io/hostname: worker-01

  tolerations:

  key: "special"

  operator: "Equal"

  value: "true"

  effect: "NoSchedule"

  containers:

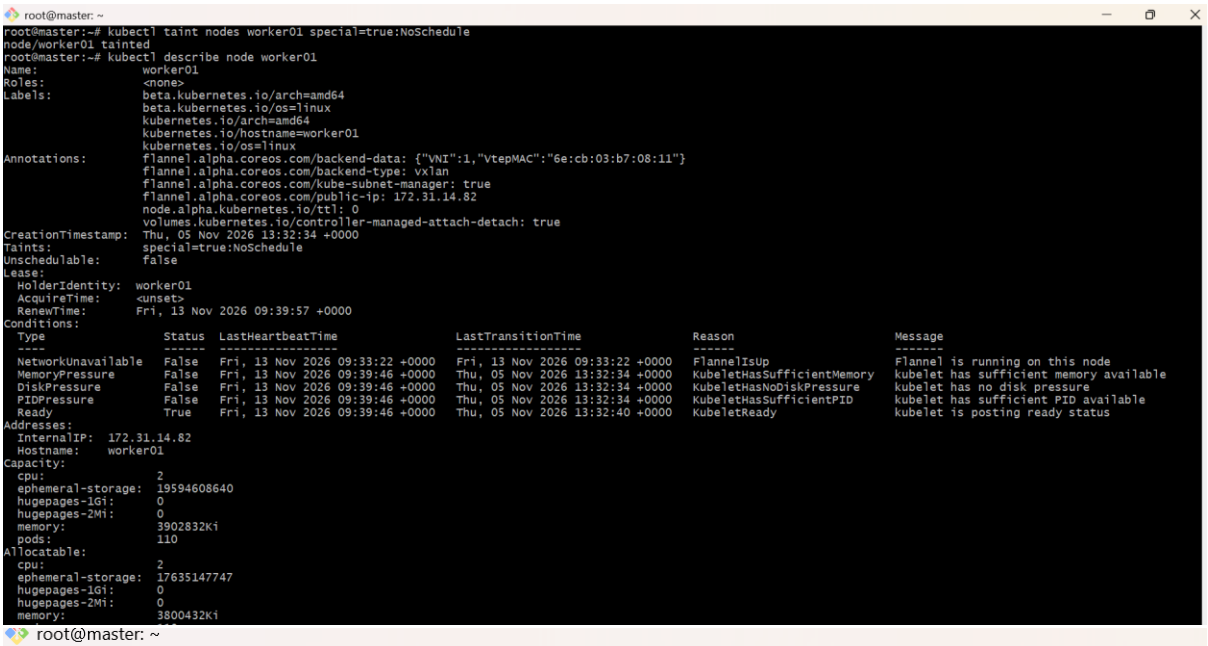
  name: nginx

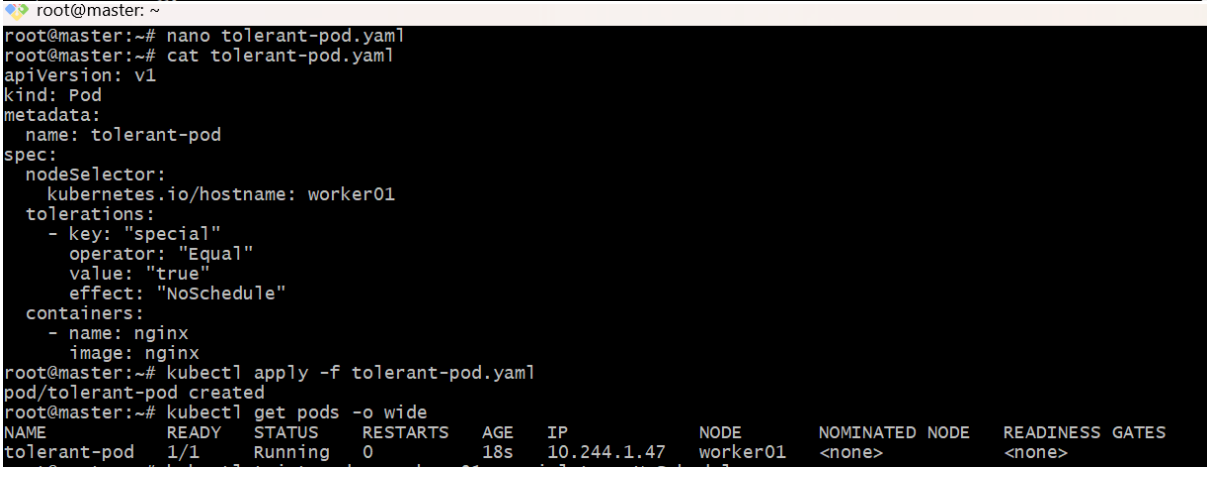
  image: nginx

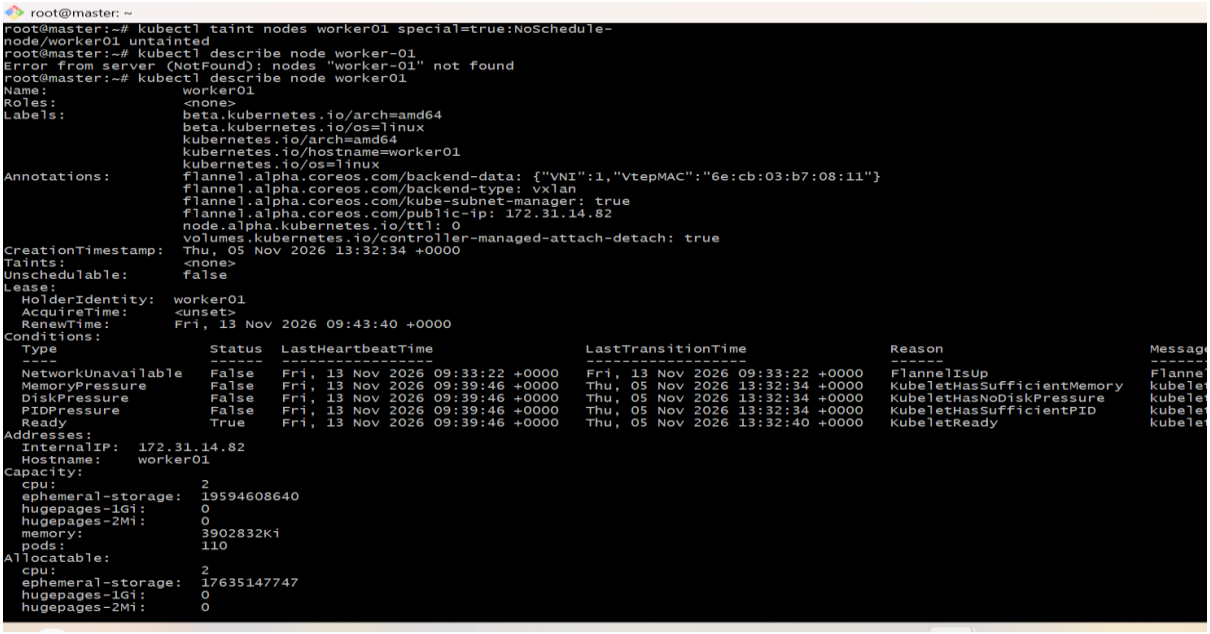
- The nodeSelector ensures the Pod is placed on Worker-01. The toleration allows it to run despite the matching taint.
- **Step 4: Create the Pod**

kubectl apply -f tolerant-pod.yaml

- **Step 5: Verify the Pod**
  kubectl get pods -o wide
- Expected result:
  NAME READY STATUS NODE
  tolerant-pod 1/1 Running worker-01
- **Step 6: Remove the taint**
  kubectl taint nodes worker-01 special=true:NoSchedule-
- Verify:
  kubectl describe node worker-01
- The special=true:NoSchedule taint should no longer appear.







Task 3: Use NodeSelector to Schedule a Pod on a Specific Node Label a node with env=dev. Create a pod with a nodeSelector that schedules it only on the node labeled env=dev.

Objective: Label a worker node with env=dev and use nodeSelector to schedule a Pod only on that node.

- **Step 1: Add a label to Worker-01**
  kubectl label nodes worker-01 env=dev
- **Step 2: Verify the node label**
  kubectl get nodes --show-labels
- You should see:
  env=dev
- on Worker-01.
- **Step 3: Create a Pod YAML file**
  vi node-selector-pod.yaml
- Add:

```
apiVersion: v1

kind: Pod

metadata:

name: node-selector-pod

spec:

nodeSelector:

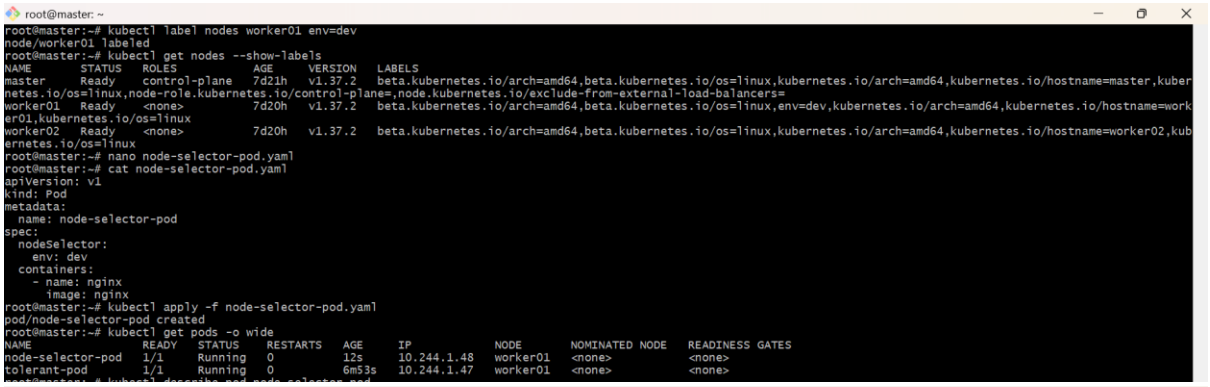
env: dev
```

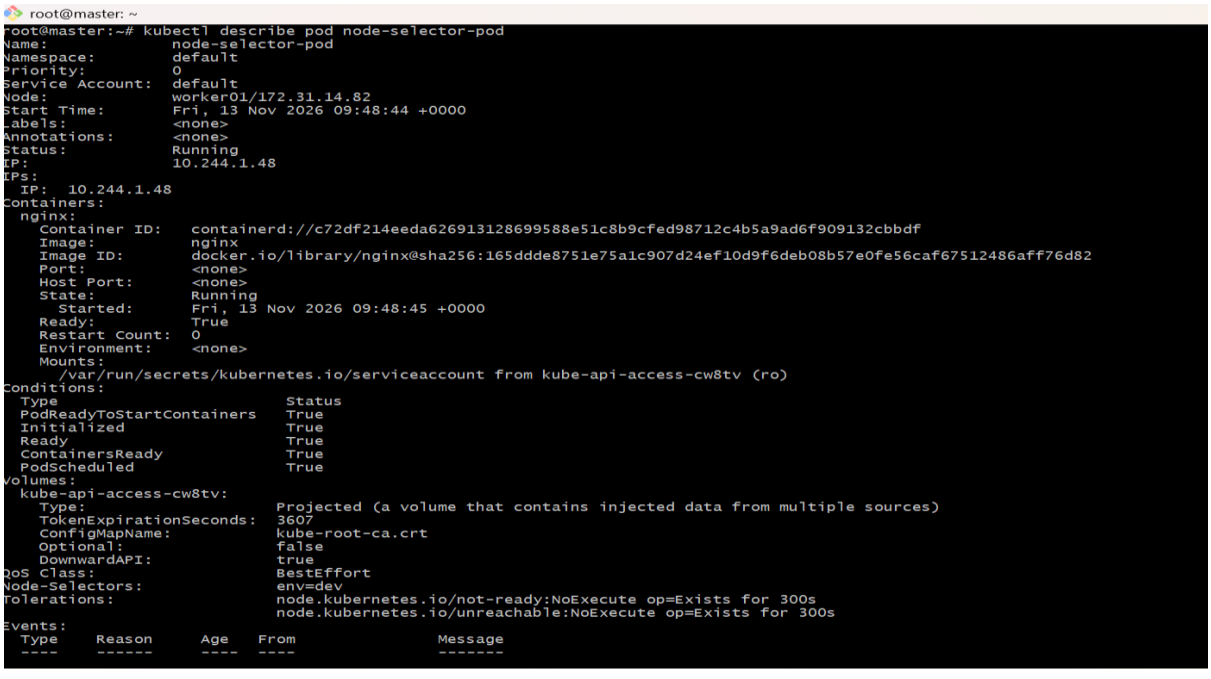
containers:

name: nginx

image: nginx

- **Step 4: Create the Pod**
  kubectl apply -f node-selector-pod.yaml
- **Step 5: Verify the Pod**
  kubectl get pods -o wide
- **Expected result:**
  NAME READY STATUS NODE
  node-selector-pod 1/1 Running worker-01
- **Step 6: Verify the scheduling details**
  kubectl describe pod node-selector-pod
- Check the Node field and confirm that the Pod is scheduled on Worker-01.
- **Step 7: Remove the label (optional cleanup)**
  kubectl label nodes worker-01 env-





Task 4: Use Node Affinity with Soft Scheduling Label a node with env=test. Create a pod with PreferredDuringSchedulingIgnoredDuringExecution node affinity, preferring to schedule it on a node labeled env=test. Remove the label and verify the pod continues to run.

Objective: Create a Pod with PreferredDuringSchedulingIgnoredDuringExecution node affinity. The scheduler should prefer a node labeled env=test. If the label is removed after the Pod starts, the Pod should continue running.

- **Step 1: Add the label to Worker-02**
  kubectl label nodes worker-02 env=test
- **Step 2: Verify the label**
  kubectl get nodes --show-labels
- **Step 3: Create a Pod YAML file**
  vi soft-affinity-pod.yaml
- Add:

  apiVersion: v1

  kind: Pod

  metadata:

  name: soft-affinity-pod

  spec:

  affinity:

  nodeAffinity:

  preferredDuringSchedulingIgnoredDuringExecution:

  weight: 1

  preference:

  matchExpressions:

  key: env

  operator: In

  values:

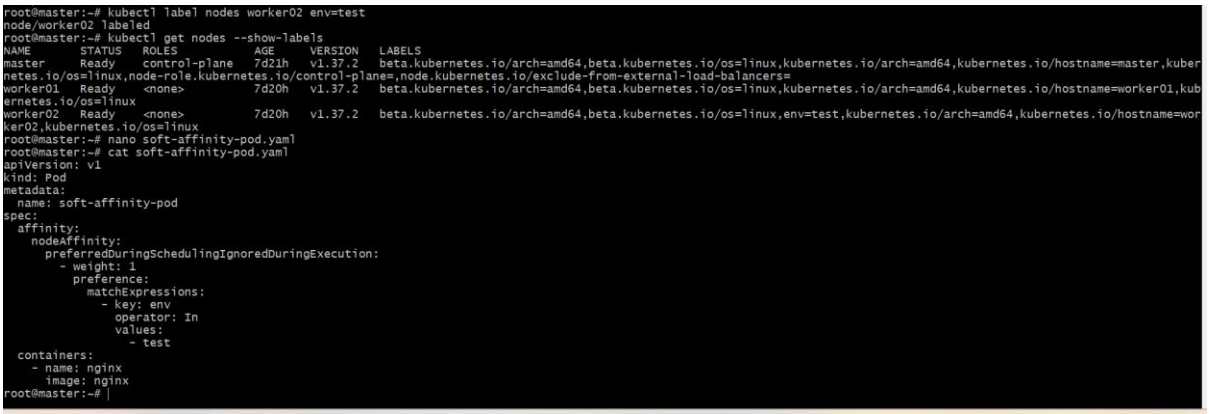
  test

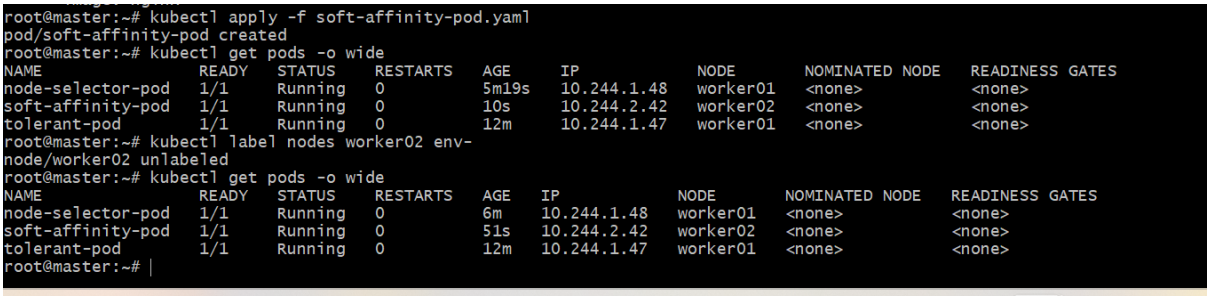
  containers:

name: nginx

image: nginx

- Explanation:
- preferredDuringSchedulingIgnoredDuringExecution means the scheduler prefers nodes matching the specified label.
- weight: 1 assigns a preference weight to the rule.
- operator: In checks whether the node label value matches test.
- If no matching node exists, the scheduler can use another eligible node.
- IgnoredDuringExecution means removing the label after scheduling does not evict the running Pod.
- **Step 4: Create the Pod**
  kubectl apply -f soft-affinity-pod.yaml
- **Step 5: Verify the Pod**
  kubectl get pods -o wide
- If Worker-02 is eligible and available, the Pod should preferably be scheduled there.
- **Step 6: Remove the label from Worker-02**
  kubectl label nodes worker-02 env-
- **Step 7: Verify that the Pod is still running**
  kubectl get pods -o wide
- **Expected result:**
  NAME READY STATUS NODE
  soft-affinity-pod 1/1 Running worker-02
- The Pod continues running even though the env=test label has been removed.





Task 5: Implement Node Affinity with Hard Scheduling Create a pod with RequiredDuringSchedulingIgnoredDuringExecution node affinity, ensuring it will only be scheduled on a node labeled env=prod. Verify the pod cannot be scheduled if no node has the env=prod label.

Objective: Create a Pod with RequiredDuringSchedulingIgnoredDuringExecution node affinity. The Pod must be scheduled only on a node labeled env=prod.

- **Step 1: Ensure no worker has the env=prod label**
  kubectl get nodes --show-labels
- If any node already has env=prod, remove that label for this test:
  kubectl label nodes worker-01 env-
  kubectl label nodes worker-02 env-
- Only run the removal commands for nodes that actually have the env label.
- **Step 2: Create a Pod YAML file**
  vi hard-affinity-pod.yaml
- Add:

  apiVersion: v1

  kind: Pod

  metadata:

  name: hard-affinity-pod

  spec:

  affinity:

  nodeAffinity:

  requiredDuringSchedulingIgnoredDuringExecution:

  nodeSelectorTerms:

  matchExpressions:

  key: env

operator: In

values:

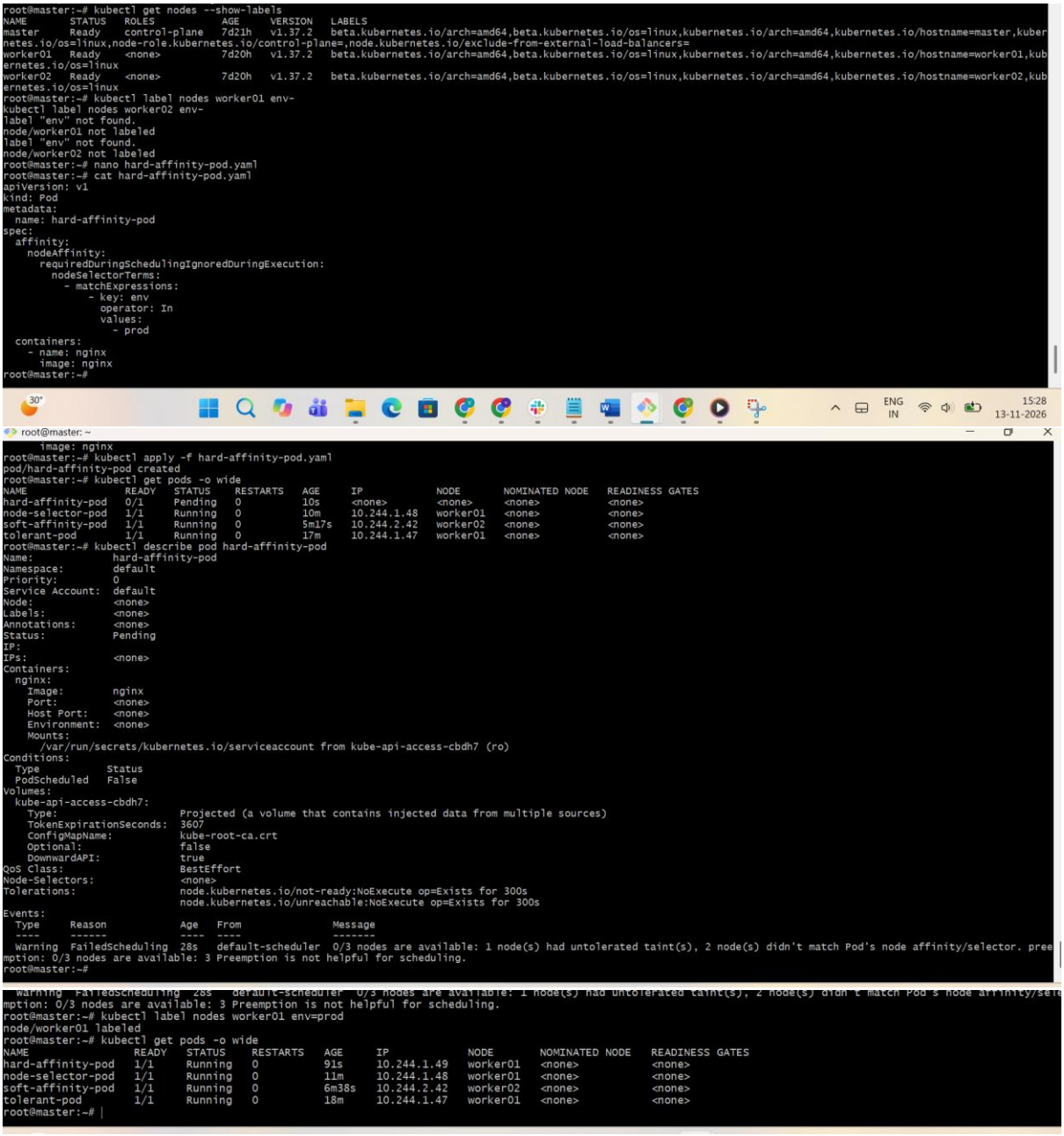
prod

containers:

name: nginx

image: nginx

- Explanation:
- requiredDuringSchedulingIgnoredDuringExecution defines a mandatory scheduling condition.
- The node must have the label env=prod.
- If no eligible node has that label, the Pod cannot be scheduled.
- Removing the label after a Pod is already running does not evict it.
- **Step 3: Create the Pod**
  kubectl apply -f hard-affinity-pod.yaml
- **Step 4: Verify the Pod status**
  kubectl get pods -o wide
- **Expected output:**
  NAME READY STATUS NODE
  hard-affinity-pod 0/1 Pending <none>
- **Step 5: Check the scheduling reason**
  kubectl describe pod hard-affinity-pod
- Under Events, you should see a scheduling message indicating that no node matches the Pod's node affinity or selector.
- **Step 6: Add the required label to Worker-01**
  kubectl label nodes worker-01 env=prod
- **Step 7: Verify the Pod again**
  kubectl get pods -o wide
- The Pod should now be scheduled on Worker-01 and move to Running status, provided the node has sufficient resources.
- **Step 8: Remove the label (optional)**
  kubectl label nodes worker-01 env-
- The already-running Pod should continue running because the rule uses IgnoredDuringExecution.



Task 6: Taint a Node and Use NoExecute with Toleration Seconds Taint a node with special=true:NoExecute. Create a pod with a tolerationSeconds field (e.g., 60 seconds) and observe it gets evicted after 60 seconds on the tainted node.

Objective: Apply a special=true:NoExecute taint to a node and create a Pod with tolerationSeconds: 60. The Pod should be evicted after tolerating the taint for 60 seconds.

- **Step 1: Create a Pod YAML file**
  vi noexecute-pod.yaml
- Add:

apiVersion: v1

kind: Pod

metadata:

name: noexecute-pod

spec:

nodeSelector:

kubernetes.io/hostname: worker-02

tolerations:

key: "special"

operator: "Equal"

value: "true"

effect: "NoExecute"

tolerationSeconds: 60

containers:

name: nginx

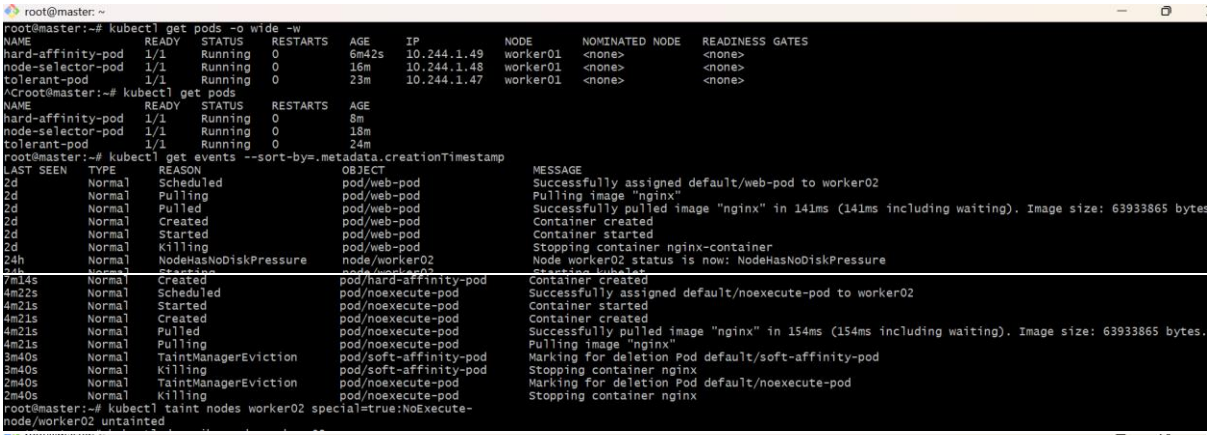
image: nginx

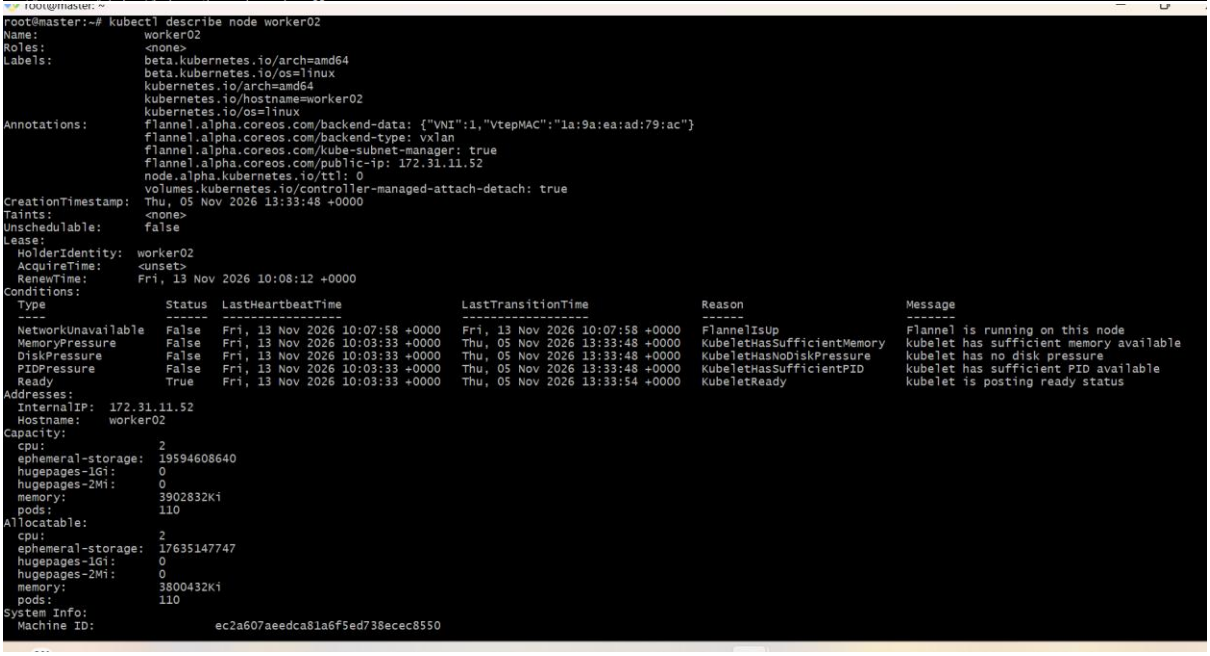
- **Step 2: Create the Pod**
  kubectl apply -f noexecute-pod.yaml
- **Step 3: Verify the Pod is running on Worker-02**
  kubectl get pods -o wide
- Confirm that the Pod is in Running status and assigned to worker-02.
- **Step 4: Apply the NoExecute taint**
  kubectl taint nodes worker-02 special=true:NoExecute
- **Step 5: Verify the taint**
  kubectl describe node worker-02
- Under Taints, verify:
  special=true:NoExecute
- **Step 6: Observe the Pod**
  kubectl get pods -o wide -w
- The Pod should remain running temporarily because it has a matching toleration with tolerationSeconds: 60.
- After approximately 60 seconds, Kubernetes will evict the Pod.
- Press Ctrl+C to stop watching.
- **Step 7: Verify the Pod eviction**
  kubectl get pods

- If the Pod was created directly using kind: Pod, it will normally disappear after eviction.
- You can also check the events:
  kubectl get events --sort-by=.metadata.creationTimestamp
- **Step 8: Remove the NoExecute taint**
  kubectl taint nodes worker-02 special=true:NoExecute-
- Verify:
  kubectl describe node worker-02
- The taint should no longer appear.

```
root@master:~# nano noexecute-pod.yaml
root@master:~# cat noexecute-pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: noexecute-pod
spec:
  nodeSelector:
    kubernetes.io/hostname: worker02
  tolerations:
    - key: "special"
      operator: "Equal"
      value: "true"
      effect: "NoExecute"
      tolerationSeconds: 60
  containers:
    - name: nginx
      image: nginx
root@master:~# kubectl apply -f noexecute-pod.yaml
pod/noexecute-pod created
root@master:~# kubectl get pods -o wide
NAME               READY   STATUS    RESTARTS   AGE     IP            NODE       NOMINATED NODE   READINESS GATES
hard-affinity-pod  1/1     Running   0          4m27s   10.244.1.49   worker01   <none>           <none>
node-selector-pod  1/1     Running   0          14m     10.244.1.48   worker01   <none>           <none>
noexecute-pod      1/1     Running   0          16s     10.244.2.43   worker02   <none>           <none>
soft-affinity-pod  1/1     Running   0          9m34s   10.244.2.42   worker02   <none>           <none>
tolerant-pod       1/1     Running   0          21m     10.244.1.47   worker01   <none>           <none>
root@master:~# kubectl taint nodes worker02 special=true:NoExecute
node/worker02 tainted
root@master:~#
```

```
root@master: ~
root@master:~# kubectl describe node worker02
Name:               worker02
Roles:              <none>
Labels:             beta.kubernetes.io/arch=amd64
                    beta.kubernetes.io/os=linux
                    kubernetes.io/arch=amd64
                    kubernetes.io/hostname=worker02
                    kubernetes.io/os=linux
Annotations:        flannel.alpha.coreos.com/backend-data: {"VNI":1,"VtepMAC":"1a:9a:ea:ad:79:ac
                    flannel.alpha.coreos.com/backend-type: vxlan
                    flannel.alpha.coreos.com/kube-subnet-manager: true
                    flannel.alpha.coreos.com/public-ip: 172.31.11.52
                    node.alpha.kubernetes.io/ttl: 0
                    volumes.kubernetes.io/controller-managed-attach-detach: true
CreationTimestamp:  Thu, 05 Nov 2026 13:33:48 +0000
Taints:             special=true:NoExecute
Unschedulable:      false
Lease:
  HolderIdentity:  worker02
  AcquireTime:     <unset>
  RenewTime:       Fri, 13 Nov 2026 10:04:29 +0000
Conditions:
  Type                 Status  LastHeartbeatTime                 LastTransitionTime
  ----                 ------  -----------------                 ------------------
  NetworkUnavailable   False   Fri, 13 Nov 2026 09:33:23 +0000   Fri, 13 Nov 2026 09:33:23 +0000
  MemoryPressure       False   Fri, 13 Nov 2026 10:03:33 +0000   Thu, 05 Nov 2026 13:33:48 +0000
  DiskPressure         False   Fri, 13 Nov 2026 10:03:33 +0000   Thu, 05 Nov 2026 13:33:48 +0000
  PIDPressure          False   Fri, 13 Nov 2026 10:03:33 +0000   Thu, 05 Nov 2026 13:33:48 +0000
  Ready                True    Fri, 13 Nov 2026 10:03:33 +0000   Thu, 05 Nov 2026 13:33:54 +0000
Addresses:
  InternalIP:  172.31.11.52
  Hostname:    worker02
Capacity:
  cpu:                2
  ephemeral-storage:  19594608640
  hugepages-1Gi:      0
  hugepages-2Mi:      0
  memory:             3902832Ki
  pods:               110
Allocatable:
  cpu:                2
  ephemeral-storage:  17635147747
  hugepages-1Gi:      0
  hugepages-2Mi:      0
  memory:             3800432Ki
  pods:               110
System Info:
  Machine ID:                 ec2a607aeedca81a6f5ed738ecec8550
```


root@master: ~
root@master:~# kubectl get pods -o wide -w
NAME READY STATUS RESTARTS AGE IP NODE NOMINATED NODE READINESS GATES
hard-affinity-pod 1/1 Running 0 6m42s 10.244.1.49 worker01 <none> <none>
node-selector-pod 1/1 Running 0 16m 10.244.1.48 worker01 <none> <none>
tolerant-pod 1/1 Running 0 23m 10.244.1.47 worker01 <none> <none>
^Croot@master:~# kubectl get pods
NAME READY STATUS RESTARTS AGE
hard-affinity-pod 1/1 Running 0 8m
node-selector-pod 1/1 Running 0 18m
tolerant-pod 1/1 Running 0 24m
root@master:~# kubectl get events --sort-by=.metadata.creationTimestamp
LAST SEEN TYPE REASON OBJECT MESSAGE
2d Normal Scheduled pod/web-pod Successfully assigned default/web-pod to worker02
2d Normal Pulling pod/web-pod Pulling image "nginx"
2d Normal Pulled pod/web-pod Successfully pulled image "nginx" in 141ms (141ms including waiting). Image size: 63933865 bytes
2d Normal Created pod/web-pod Container created
2d Normal Started pod/web-pod Container started
2d Normal Killing pod/web-pod Stopping container nginx-container
24h Normal NodeHasNoDiskPressure node/worker02 Node worker02 status is now: NodeHasNoDiskPressure
24h Normal Starting node/worker02 Starting kubelet.
7m14s Normal Created pod/hard-affinity-pod Container created
4m22s Normal Scheduled pod/noexecute-pod Successfully assigned default/noexecute-pod to worker02
4m21s Normal Started pod/noexecute-pod Container started
4m21s Normal Created pod/noexecute-pod Container created
4m21s Normal Pulled pod/noexecute-pod Successfully pulled image "nginx" in 154ms (154ms including waiting). Image size: 63933865 bytes.
4m21s Normal Pulling pod/noexecute-pod Pulling image "nginx"
3m40s Normal TaintManagerEviction pod/soft-affinity-pod Marking for deletion Pod default/soft-affinity-pod
3m40s Normal Killing pod/soft-affinity-pod Stopping container nginx
2m40s Normal TaintManagerEviction pod/noexecute-pod Marking for deletion Pod default/noexecute-pod
2m40s Normal Killing pod/noexecute-pod Stopping container nginx
root@master:~# kubectl taint nodes worker02 special=true:NoExecute-
node/worker02 untainted


root@master:~# kubectl describe node worker02
Name: worker02
Roles: <none>
Labels: beta.kubernetes.io/arch=amd64
beta.kubernetes.io/os=linux
kubernetes.io/arch=amd64
kubernetes.io/hostname=worker02
kubernetes.io/os=linux
Annotations: flannel.alpha.coreos.com/backend-data: {"VNI":1,"VtepMAC":"1a:9a:ea:ad:79:ac"}
flannel.alpha.coreos.com/backend-type: vxlan
flannel.alpha.coreos.com/kube-subnet-manager: true
flannel.alpha.coreos.com/public-ip: 172.31.11.52
node.alpha.kubernetes.io/ttl: 0
volumes.kubernetes.io/controller-managed-attach-detach: true
CreationTimestamp: Thu, 05 Nov 2026 13:33:48 +0000
Taints: <none>
Unschedulable: false
Lease:
HolderIdentity: worker02
AcquireTime: <unset>
RenewTime: Fri, 13 Nov 2026 10:08:12 +0000
Conditions:
Type Status LastHeartbeatTime LastTransitionTime Reason Message
NetworkUnavailable False Fri, 13 Nov 2026 10:07:58 +0000 Fri, 13 Nov 2026 10:07:58 +0000 FlannelIsUp Flannel is running on this node
MemoryPressure False Fri, 13 Nov 2026 10:03:33 +0000 Thu, 05 Nov 2026 13:33:48 +0000 KubeletHasSufficientMemory kubelet has sufficient memory available
DiskPressure False Fri, 13 Nov 2026 10:03:33 +0000 Thu, 05 Nov 2026 13:33:48 +0000 KubeletHasNoDiskPressure kubelet has no disk pressure
PIDPressure False Fri, 13 Nov 2026 10:03:33 +0000 Thu, 05 Nov 2026 13:33:48 +0000 KubeletHasSufficientPID kubelet has sufficient PID available
Ready True Fri, 13 Nov 2026 10:03:33 +0000 Thu, 05 Nov 2026 13:33:54 +0000 KubeletReady kubelet is posting ready status
Addresses:
InternalIP: 172.31.11.52
Hostname: worker02
Capacity:
cpu: 2
ephemeral-storage: 19594608640
hugepages-1Gi: 0
hugepages-2Mi: 0
memory: 3902832Ki
pods: 110
Allocatable:
cpu: 2
ephemeral-storage: 17635147747
hugepages-1Gi: 0
hugepages-2Mi: 0
memory: 3800432Ki
pods: 110
System Info:
Machine ID: ec2a607aeedca81a6f5ed738ecec8550